

Corrected calculations for network paper

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PRELIMINARIES

These are preliminary identities that can be derived by using the finite series identities in PRX. I have derived the following for $|m| < N$ and $m \neq 0$

$$\frac{1}{N} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} = 1 \quad (1)$$

Another identity exploits the fact that

$$\begin{aligned} [1 - (-1)^k] \sin \left[\frac{|m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right] &= \frac{1}{2} \left\{ \cos \left[\frac{(|m| - 1)\pi k}{N} \right] - \cos \left[\frac{(|m| + 1)\pi k}{N} \right] \right. \\ &\quad \left. - \frac{1}{2} \left[\cos \left[\frac{(N - |m| + 1)\pi k}{N} \right] + \cos \left[\frac{(N + |m| - 1)\pi k}{N} \right] - \cos \left[\frac{(N - |m| - 1)\pi k}{N} \right] - \cos \left[\frac{(N + |m| + 1)\pi k}{N} \right] \right] \right\} \end{aligned}$$

for a generic σ we have

$$\begin{aligned} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{\sigma - \cos \left[\frac{\pi k}{N} \right]} &= \frac{1}{2} \sum_{k=1}^N \frac{\cos \left[\frac{(|m| - 1)\pi k}{N} \right] - \cos \left[\frac{(|m| + 1)\pi k}{N} \right]}{\sigma - \cos \left[\frac{\pi k}{N} \right]} \\ &\quad - \frac{1}{4} \sum_{k=1}^{N-1} \frac{\cos \left[\frac{(N - |m| + 1)\pi k}{N} \right] + \cos \left[\frac{(N + |m| - 1)\pi k}{N} \right] - \cos \left[\frac{(N - |m| - 1)\pi k}{N} \right] - \cos \left[\frac{(N + |m| + 1)\pi k}{N} \right]}{\sigma - \cos \left[\frac{\pi k}{N} \right]} \\ &= \frac{N}{2(\sigma^2 - 1)U_{N-1}(\sigma)} \{T_{N-|m|+1}(\sigma) - T_{N-|m|-1}(\sigma) - T_{|m|-1}(\sigma) + T_{|m|+1}(\sigma)\} \end{aligned} \quad (2)$$

If $\sigma = \cos \left[\frac{2\pi r}{N} \right]$ one needs to use de l'Hospital rule to evaluate the expression above, which gives

$$\begin{aligned} \lim_{\sigma \rightarrow \cos \left[\frac{2\pi r}{N} \right]} \frac{N}{2(\sigma^2 - 1)U_{N-1}(\sigma)} \{T_{N-|m|+1}(\sigma) - T_{N-|m|-1}(\sigma) - T_{|m|-1}(\sigma) + T_{|m|+1}(\sigma)\} \\ = \cos \left[\frac{2\pi r|m|}{N} \right], \end{aligned} \quad (3)$$

in other words we have the general identity for $0 < |m| < N$ and $r = 1, \dots, N-1$,

$$\frac{1}{N} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{\cos \left[\frac{2\pi r}{N} \right] - \cos \left[\frac{\pi k}{N} \right]} = \cos \left[\frac{2\pi r|m|}{N} \right], \quad (4)$$

The third relevant identity for $|m| \leq N$ and integer k is given by

$$\frac{1}{N} \sum_{r=1}^{N-1} \frac{\cos \left[\frac{2\pi r m}{N} \right]}{\cos \left[\frac{(2k-1)\pi}{N} \right] - \cos \left[\frac{2\pi r}{N} \right]} = \frac{1}{N \left\{ 1 - \cos \left[\frac{(2k-1)\pi}{N} \right] \right\}} - \frac{\sin \left[\frac{(2k-1)\pi|m|}{N} \right]}{\sin \left[\frac{(2k-1)\pi}{N} \right]} \quad (5)$$

alternatively

$$\frac{1}{N} \sum_{r=1}^{N-1} [1 - (-1)^k] \frac{\cos \left[\frac{2\pi r m}{N} \right]}{\cos \left[\frac{k\pi}{N} \right] - \cos \left[\frac{2\pi r}{N} \right]} = [1 - (-1)^k] \left\{ \frac{1}{N \left\{ 1 - \cos \left[\frac{\pi k}{N} \right] \right\}} - \frac{\sin \left[\frac{\pi k|m|}{N} \right]}{\sin \left[\frac{\pi k}{N} \right]} \right\} \quad (6)$$

PROPAGATOR IN RING LATTICE WITH ABSORPTION SITE

The occupation probability for the periodic ring lattice with arbitrary K and an absorption site at $n = m$ can be written via the general relation for any $t \geq 1$

$$W_{n_0}(n, t) = Q_{n_0}^{(k)}(n, t) - \sum_{s=1}^t \mathcal{F}_{n_0 \rightarrow m}(s) Q_m^{(k)}(n, t-s) \quad (7)$$

where $Q_r^{(k)}(n, t)$ is the propagator for the periodic ring lattice in the absence of any absorption site. Note that at $t = 0$, by construction $F_{n_0 \rightarrow m}(0) = 0$ when $n_0 \neq m$, and we have $W_{n_0}(n, 0) = Q_{n_0}^{(k)}(n, 0)$.

More explicit expressions can be obtained for the case $K = 2$, that is the nearest-neighbour lattice walk. We have that

$$Q_{n_0}^{(1)}(n, t) = \frac{1}{N} \sum_{k=0}^{N-1} \cos \left[\frac{2\pi k(n - n_0)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t, \quad (8)$$

and for $t \geq 1$

$$\mathcal{F}_{n_0}(m, t) = \frac{q}{N} \sum_{k=1}^{N-1} [1 - (-1)^k] \sin \left[\frac{|m - n_0|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right] \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^{t-1}. \quad (9)$$

with $\mathcal{F}_{n_0}(m, 0) = 0$.

Given that in Eq. (9) only odd k terms are non-zero, the first-passage probability can be written in compact form as

$$\mathcal{F}_{n_0}(m, t) = \sum_{k=1}^{\frac{N}{2}} f_k(n_0, m) \alpha_k^{t-1}. \quad (10)$$

where

$$f_k(n_0, m) = \frac{2q}{N} \sin \left[\frac{|m - n_0|\pi(2k-1)}{N} \right] \sin \left[\frac{\pi(2k-1)}{N} \right] \quad (11)$$

and

$$\alpha_k = 1 - q + q \cos \left[\frac{\pi(2k-1)}{N} \right]. \quad (12)$$

The generating function of Eq. (10) in compact form can be written as

$$\tilde{\mathcal{F}}_{n_0}(m, z) = z \sum_{k=1}^{\frac{N}{2}} \frac{f_k(n_0, m)}{1 - z\alpha_k} = \frac{U_{N-|m-n_0|-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) + U_{|m-n_0|-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right)}{U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right)}. \quad (13)$$

where $U_r(y)$ is the Chebyshev polynomial of the second kind.

Also we note that using Eq. (1) one can show that the cumulative absorption probability is

$$\sum_{s=1}^t \mathcal{F}_{n_0}(m, s) = 1 - \frac{1}{N} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|n_0 - m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t,$$

which gives the survival probability

$$R_{n_0}(t) = 1 - \sum_{s=0}^t \mathcal{F}_{n_0}(m, s) = \frac{1}{N} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|n_0 - m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t.$$

Using the above expressions in Eq. (7) and evaluating the time summation, we obtain

$$\begin{aligned}
W_{n_0}(n, t) = & \frac{1}{N} \sum_{k=1}^{N-1} \cos \left[\frac{2\pi k(n-n_0)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t \\
& + \frac{1}{N^2} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|n_0-m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \\
& + \frac{1}{N^2} \sum_{k=1}^{N-1} \sum_{r=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|n_0-m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right] \cos \left[\frac{2\pi r(n-m)}{N} \right]}{\cos \left[\frac{\pi k}{N} \right] - \cos \left[\frac{2\pi r}{N} \right]} \\
& \times \left\{ \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t - \left[1 - q + q \cos \left(\frac{2\pi r}{N} \right) \right]^t \right\}
\end{aligned} \tag{14}$$

Using the identities (4) and (6), we reduce the double summation to single summations as follows

$$\begin{aligned}
& \sum_{k=1}^{N-1} \sum_{r=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|n_0-m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right] \cos \left[\frac{2\pi r(n-m)}{N} \right]}{\cos \left[\frac{\pi k}{N} \right] - \cos \left[\frac{2\pi r}{N} \right]} \left\{ \left[1 - q + q \cos \left(\frac{2\pi r}{N} \right) \right]^t - \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \right\} \\
& = -N \sum_{r=1}^{N-1} \cos \left[\frac{2\pi r|n_0-m|}{N} \right] \cos \left[\frac{2\pi r(n-m)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi r}{N} \right) \right]^t \\
& - \sum_{k=1}^{N-1} [1 - (-1)^k] \left\{ \frac{\sin \left[\frac{|n_0-m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} - N \sin \left[\frac{|n_0-m|\pi k}{N} \right] \sin \left[\frac{|n-m|\pi k}{N} \right] \right\} \\
& \times \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \\
& = -N \sum_{r=1}^{N-1} \cos \left[\frac{2\pi r|n_0-m|}{N} \right] \cos \left[\frac{2\pi r(n-m)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi r}{N} \right) \right]^t \\
& - \sum_{k=1}^{N-1} [1 - (-1)^k] \left\{ \frac{\sin \left[\frac{|n_0-m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} \right. \\
& \left. - N \sin \left[\frac{|n_0-m|\pi k}{N} \right] \sin \left[\frac{|n-m|\pi k}{N} \right] \right\} \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t
\end{aligned} \tag{15}$$

and we have

$$\begin{aligned}
W_{n_0}(n, t) = & \frac{1}{N} \sum_{k=1}^{N-1} \cos \left[\frac{2\pi k(n-n_0)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t \\
& - \frac{1}{N} \sum_{k=1}^{N-1} \cos \left[\frac{2\pi k(n-m)}{N} \right] \cos \left[\frac{2\pi k(n_0-m)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t \\
& + \frac{1}{N} \sum_{k=1}^{N-1} [1 - (-1)^k] \sin \left[\frac{\pi k|n_0-m|}{N} \right] \sin \left[\frac{\pi k|n-m|}{N} \right] \left\{ 1 - q + q \cos \left[\frac{\pi k}{N} \right] \right\}^t.
\end{aligned} \tag{16}$$

As $\sum_{k=1}^{N-1} \cos \left[\frac{2\pi k(n-m)}{N} \right] \cos \left[\frac{2\pi k(n_0-m)}{N} \right] = -1$ and $\sum_{k=1}^{N-1} [1 - (-1)^k] \sin \left[\frac{\pi k|n_0-m|}{N} \right] \sin \left[\frac{\pi k|n-m|}{N} \right] = 0$, one can show that $W_{n_0}(n, 0) = \delta_{n, n_0}$ for $n, n_0 \neq m$. Furthermore, for the killed propagator W , when either $n = m$ or $n_0 = m$ one has $W_{n_0}(n, t) = 0$, and it is also easy to see that $W_{n_0}(n, t \rightarrow \infty) = 0$, which means that all random walks eventually reach the absorbing site $n = m$.

We can rewrite Eq. (16) as

$$W_{n_0}(n, t) = \sum_{k=1}^{N-1} G_k^{(1)}(n_0, n, m) \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t + \sum_{k=1}^{N-1} G_k^{(2)}(n_0, n, m) \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \tag{17}$$

where m is the absorbing site, and with

$$\begin{aligned} G_k^{(1)}(n_0, n, m) &= \frac{\cos \left[\frac{2\pi k(n-n_0)}{N} \right] - \cos \left[\frac{2\pi k(n-m)}{N} \right] \cos \left[\frac{2\pi k(n_0-m)}{N} \right]}{N} \\ G_k^{(2)}(n_0, n, m) &= \frac{[1 - (-1)^k] \sin \left[\frac{\pi k|n_0-m|}{N} \right] \sin \left[\frac{\pi k|n-m|}{N} \right]}{N} \end{aligned} \quad (18)$$

In terms of generating functions one has

$$\widetilde{W}_{n_0}(n, z) = \sum_{k=1}^{N-1} \frac{G_k^{(1)}(n_0, n, m)}{1 - z [1 - q + q \cos \left(\frac{2\pi k}{N} \right)]} + \sum_{k=1}^{N-1} \frac{G_k^{(2)}(n_0, n, m)}{1 - z [1 - q + q \cos \left(\frac{\pi k}{N} \right)]} \quad (19)$$

or the equivalent form

$$\widetilde{W}_{n_0}(n, z) = \sum_{k=1}^{N-1} \frac{G_k^{(1)}(n_0, n, m)}{1 - z\gamma_k} + \sum_{k=1}^{\frac{N}{2}} \frac{g_k^{(2)}(n_0, n, m)}{1 - z\alpha_k}, \quad (20)$$

and its time-dependent version

$$W_{n_0}(n, t) = \sum_{k=1}^{N-1} G_k^{(1)}(n_0, n, m) \gamma_k^t + \sum_{k=1}^{\frac{N}{2}} g_k^{(2)}(n_0, n, m) \alpha_k^t, \quad (21)$$

where α_k is defined in Eq. (12),

$$\gamma_k = 1 - q + q \cos \left(\frac{2\pi k}{N} \right), \quad (22)$$

and

$$g_k^{(2)}(n_0, n, m) = \frac{2}{N} \sin \left[\frac{\pi(2k-1)|n_0-m|}{N} \right] \sin \left[\frac{\pi(2k-1)|n-m|}{N} \right]. \quad (23)$$

The associated expressions in terms of Chebyshev polynomials of the first and second kind, respectively, $T_r(y)$ and $U_r(y)$, are

$$\begin{aligned} \sum_{k=1}^{N-1} \frac{G_k^{(1)}(n_0, n, m)}{1 - z [1 - q + q \cos \left(\frac{2\pi k}{N} \right)]} &= \left\{ 2T_{N-|n-n_0|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) + 2T_{|n-n_0|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right. \\ &\quad - T_{N-|n_0-m|-|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - T_{|n_0-m|-|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \\ &\quad \left. - T_{N-[|n_0-m|+|n-m| \pmod{N}]} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - T_{|n_0-m|+|n-m| \pmod{N}} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right\} \\ &\quad \times \left[2zq \left\{ \left[1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right) \right]^2 - 1 \right\} U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right]^{-1}, \end{aligned} \quad (24)$$

and

$$\begin{aligned} \sum_{k=1}^{N-1} \frac{G_k^{(2)}(n_0, n, m)}{1 - z [1 - q + q \cos \left(\frac{\pi k}{N} \right)]} &= \left\{ T_{N-|n_0-m|-|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) + T_{N-|N-|n_0-m|-|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right. \\ &\quad \left. - T_{|N-|n_0-m|-|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - T_{|n_0-m|-|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right\} \\ &\quad \times \left[2zq \left\{ \left[1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right) \right]^2 - 1 \right\} U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right]^{-1}, \end{aligned} \quad (25)$$

When $n = n_0$ Eq. (17) reduces to

$$W_n(n, t) = \frac{1}{N} \sum_{k=1}^{N-1} \left\{ \left(1 - \cos^2 \left[\frac{2\pi k(n-m)}{N} \right] \right) \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t + [1 - (-1)^k] \sin^2 \left[\frac{\pi k|n-m|}{N} \right] \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \right\}, \quad (26)$$

whose generating function is given by the compact expression

$$\begin{aligned} \widetilde{W}_n(n, z) = & \left\{ 2T_N \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) + T_{N-|N-2|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - T_{|N-2|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right. \\ & \left. - T_{N-[2|n-m| \pmod{N}]} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - T_{[2|n-m| \pmod{N}]} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right\} \\ & \left\{ 2zq \left\{ \left[1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right) \right]^2 - 1 \right\} U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right\}^{-1} \end{aligned} \quad (27)$$

Note that using Eqs. (19) with $n = n_0$ and (27) when $|n-m| = N/2$ with N even, one can show setting $z = 1$ the following identities

$$\sum_{k=1}^{N-1} \frac{1 - (-1)^k}{1 - \cos \left(\frac{\pi k}{N} \right)} = \frac{N^2}{2}, \quad \sum_{k=1}^{N-1} \frac{1}{1 - \cos \left(\frac{\pi k}{N} \right)} = \frac{N^2 - 1}{3}, \quad \sum_{k=1}^{N-1} \frac{(-1)^k}{1 - \cos \left(\frac{\pi k}{N} \right)} = -\frac{N^2 + 2}{6}.$$

For the special case $|n-m| = N/2$ with N even Eq. (27) becomes

$$\widetilde{W}_n(n, z) = \frac{T_N \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - 1}{zq \left\{ \left[1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right) \right]^2 - 1 \right\} U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right)} = \frac{2 \left[U_{\frac{N}{2}-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right]^2}{zq U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right)} \quad (28)$$

PROPAGATOR WITH ONE SHORTCUT AND ONE ABSORBING SITE

We start from the general propagator with a shortcut from $n = u$ to site $n = v$ and we have

$$\widetilde{S}_{n_0}(n, z) = \widetilde{P}_{n_0}(n, z) - \frac{z\beta(1-q)\widetilde{P}_{n_0}(u, z) \left[\widetilde{P}_u(n, z) - \widetilde{P}_v(n, z) \right]}{1 + z\beta(1-q) \left[\widetilde{P}_u(u, z) - \widetilde{P}_v(u, z) \right]}, \quad (29)$$

where $\widetilde{P}_r(n, z)$ is the ring propagator generating function without shortcuts. If the ring also has an absorbing site, the modularity of the defect technique allows to replace the (probability conserving) propagator $\widetilde{P}_r(n, z)$ with the propagator $\widetilde{W}_r(n, z)$ in Eq. (17)

$$\widetilde{S}_{n_0}(n, z) = \widetilde{W}_{n_0}(n, z) - \frac{z\beta(1-q)\widetilde{W}_{n_0}(u, z) \left[\widetilde{W}_u(n, z) - \widetilde{W}_v(n, z) \right]}{1 + z\beta(1-q) \left[\widetilde{W}_u(u, z) - \widetilde{W}_v(u, z) \right]}. \quad (30)$$

When the absorbing site coincides with the shortcut site, i.e. $m = v$, $\widetilde{W}_v(n, z) = 0$ and Eq. (30) simplifies further to

$$\widetilde{S}_{n_0}(n, z) = \widetilde{W}_{n_0}(n, z) - \frac{z\beta(1-q)\widetilde{W}_{n_0}(u, z)\widetilde{W}_u(n, z)}{1 + z\beta(1-q)\widetilde{W}_u(u, z)}. \quad (31)$$

After summing Eq. (31) over n to obtain the survival probability $\tilde{\mathcal{S}}_{n_0}(z) = [1 - \tilde{F}_{n_0}(v, z)] / (1 - z)$, one expresses the generating function of the first-passage probability to site v as

$$\tilde{F}_{n_0}(v, z) = \tilde{\mathcal{F}}_{n_0}(v, z) + \frac{z\beta(1-q)\tilde{W}_{n_0}(u, z) [1 - \tilde{\mathcal{F}}_u(v, z)]}{1 + z\beta(1-q)\tilde{W}_u(u, z)}, \quad (32)$$

or more explicitly as

$$\tilde{F}_{n_0}(v, z) = \tilde{\mathcal{F}}_{n_0}(v, z) + qz\tilde{W}_{n_0}(u, z) [1 - \tilde{\mathcal{F}}_u(v, z)] \tilde{H}_{N, |u-v|}(z), \quad (33)$$

with $\tilde{W}_{n_0}(u, z)$ and $\tilde{\mathcal{F}}_r(m, z)$ defined, respectively, in Eqs. (19) and (13), and

$$\tilde{H}_{N, |u-v|}(z) = \frac{\beta(1-q)}{q} \frac{1}{1 + z\beta(1-q)\tilde{W}_u(u, z)}. \quad (34)$$

We now consider the symmetric case $|u - v| = N/2$ with N even, and use $\tilde{W}_u(u, z)$ from Eq. (28) in Eq. (34), which allows us to simplify Eq. (34) to

$$\tilde{H}_{N, N/2}(z) = \frac{1}{\frac{q}{\beta(1-q)} + \phi(z)}, \quad (35)$$

where

$$\phi(z) = \frac{2U_{N/2-1}^2 \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1\right]\right)}{U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1\right]\right)}. \quad (36)$$

whose time-dependence is $H_{N, N/2}(0) = \beta(1-q)/q$ and for $t \geq 1$ is given by the complex integral (after the change of integration variable $z = 1/s$ and the region of integration becoming $|s| \leq 1$)

$$H_{N, N/2}(t) = \frac{1}{2\pi i} \oint_{|s| \leq 1} \frac{s^{t-1}}{\frac{q}{\beta(1-q)} + \phi\left(\frac{1}{s}\right)}. \quad (37)$$

The denominator in the integrand of Eq. (37) has real zeros s_k , which are the intersections of $\phi(1/s)$ with the horizontal line $-q/[\beta(1-q)]$. These intersections are shown graphically in Fig. 1.

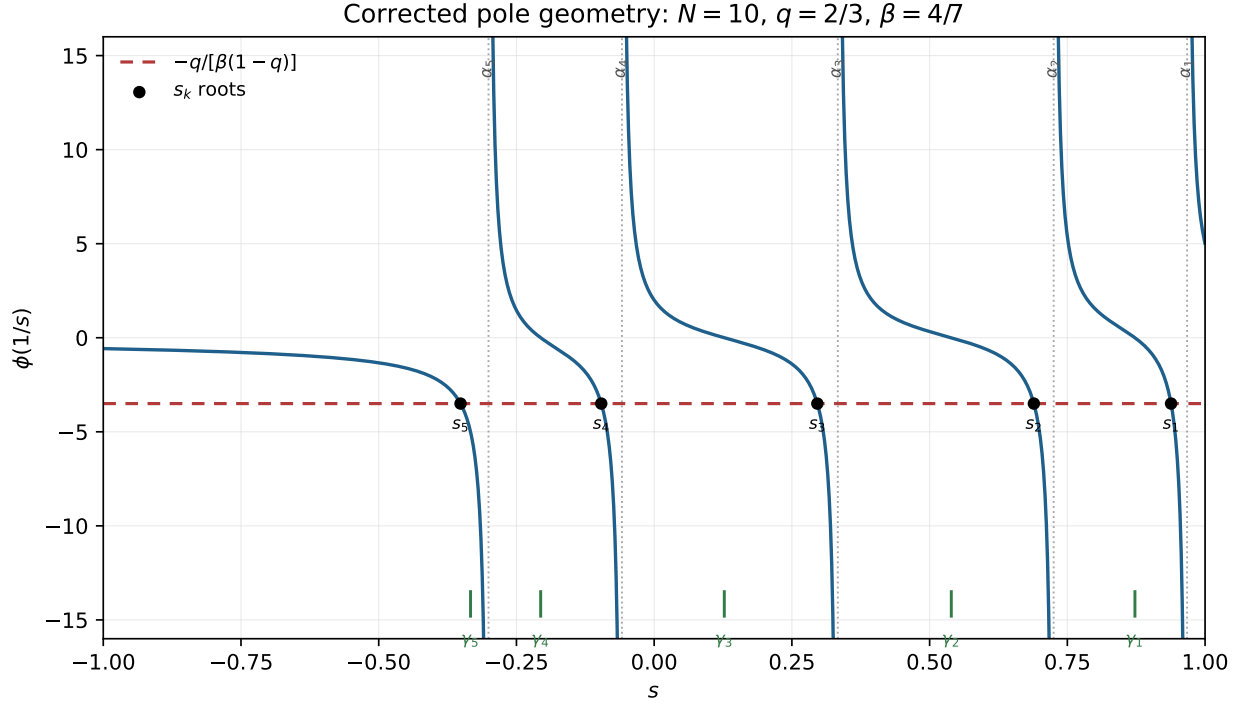


FIG. 1. Graphical representation of the poles of Eq. (35) for the case $N = 10$, $q = 2/3$, and $\beta = 4/7$. The poles are the values of s where the horizontal line $-q/[\beta(1-q)]$ intercepts $\phi(1/s)$. The asymptotes of $\phi(1/s)$ are given by the α_k , and the paired tick marks indicate the corresponding γ_k . This figure is a visualization of the H -poles; the finite-time one-peak/two-peak transition is a property of the full first-passage distribution.

It is useful to rewrite the same corrected H in a form tied directly to its denominator. Let $N = 2L$,

$$a = \frac{q}{\beta(1-q)}, \quad y = 1 + \frac{1/z - 1}{q}.$$

Then

$$\tilde{H}_{N,N/2}(z) = \frac{T_L(y)}{aT_L(y) + U_{L-1}(y)}. \quad (38)$$

If $D(y) = aT_L(y) + U_{L-1}(y)$, $D(y_k) = 0$, and $s_k = 1 - q + qy_k$, then the positive-time part of H can be written, for simple roots, as

$$H_{N,N/2}(t) = \sum_k c_k s_k^{t-1}, \quad c_k = \frac{qT_L(y_k)}{D'(y_k)}, \quad t \geq 1. \quad (39)$$

This residue form keeps the $H(0) = \beta(1-q)/q$ contribution separate from the positive-time pole expansion.

The time dependence of Eq. (33) for $|u - v| = N/2$ is therefore given by the time-convoluted expression for $t \geq 1$

$$F_{n_0}(v, t) = \mathcal{F}_{n_0}(v, t) + q \sum_{t'=0}^{t-1} W_{n_0}(u, t-1-t') \sum_{t''=0}^{t'} [\delta_{t', t''} - \mathcal{F}_u(v, t' - t'')] H_{N,N/2}(t'') \quad (40)$$

and $F_{n_0}(v, 0) = 0$.

The same result can be written more compactly at the generating-function level. Let ρ denote the folded distance

of the starting site from the absorbing target in the antipodal case. Using

$$\tilde{\mathcal{F}}_\rho(0, z) = \frac{T_{L-\rho}(y)}{T_L(y)}, \quad (41)$$

$$\tilde{W}_\rho(L, z) = \frac{U_{\rho-1}(y)}{zq T_L(y)}, \quad (42)$$

$$1 - \tilde{\mathcal{F}}_L(0, z) = \frac{T_L(y) - 1}{T_L(y)}, \quad (43)$$

$$\tilde{H}_{N,N/2}(z) = \frac{T_L(y)}{aT_L(y) + U_{L-1}(y)}, \quad (44)$$

Eq. (33) reduces to

$$\begin{aligned} \tilde{F}_\rho(0, z) &= \frac{T_{L-\rho}(y)}{T_L(y)} + \frac{U_{\rho-1}(y)[T_L(y) - 1]}{T_L(y)[aT_L(y) + U_{L-1}(y)]} \\ &= \frac{aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)}{aT_L(y) + U_{L-1}(y)}. \end{aligned} \quad (45)$$

The second equality uses

$$T_L(y)U_{L-\rho-1}(y) = T_{L-\rho}(y)U_{L-1}(y) - U_{\rho-1}(y).$$

The expanded-route check in the Appendix shows the same cancellation explicitly. Thus the long-time decay is controlled by the roots of

$$D(y) = aT_L(y) + U_{L-1}(y).$$

For simple roots $D(y_k) = 0$, $s_k = 1 - q + qy_k$,

$$F_\rho(0, t) = \sum_k B_{\rho k} s_k^{t-1}, \quad B_{\rho k} = \frac{q[aT_{L-\rho}(y_k) + U_{\rho-1}(y_k) + U_{L-\rho-1}(y_k)]}{D'(y_k)}. \quad (46)$$

Therefore, ordering the roots so that s_1 is the dominant positive pole and assuming $B_{\rho 1} \neq 0$, the corrected long-time dependence is

$$F_\rho(0, t) \sim B_{\rho 1} s_1^{t-1}, \quad t \rightarrow \infty.$$

Equivalently, the exponential decay rate is $-\log s_1$. The dependence on the shortcut strength enters through

$$a = \frac{q}{\beta(1-q)}$$

in the denominator $D(y) = aT_L(y) + U_{L-1}(y)$, and hence through the dominant root $s_1 = 1 - q + qy_1$. Thus the corrected shortcut calculation does not produce a generic $t\alpha_1^t$ long-time factor; the long-time decay is set by the dominant root of the corrected denominator. The finite-time one-peak/two-peak transition is therefore a shape question for the full first-passage distribution, not a criterion obtained from the auxiliary H -pole plot alone. In particular, for finite positive β in the corrected pole equation, the proposed crossing $s_1 = \gamma_1 = 1 - q + q \cos(2\pi/N)$ does not occur; the dominant corrected pole satisfies $\gamma_1 < s_1 < \alpha_1$.

APPENDIX: EXPANDED-ROUTE CHECK

This appendix records the same calculation in the expanded route, starting from the corrected generating-function version of the time convolution. It is included to show how the apparent $T_L(y)$ -denominators cancel once the corrected shortcut sign has been propagated.

Let $N = 2L$,

$$a = \frac{q}{\beta(1-q)}, \quad y = 1 + \frac{1/z - 1}{q}, \quad D(y) = aT_L(y) + U_{L-1}(y).$$

For the antipodal shortcut geometry, Eq. (33) can be written as

$$\tilde{F}_\rho(0, z) = \tilde{\mathcal{F}}_\rho(0, z) + qz \tilde{W}_\rho(L, z) \left[1 - \tilde{\mathcal{F}}_L(0, z) \right] \tilde{H}_{N, N/2}(z). \quad (47)$$

Substituting

$$\begin{aligned} \tilde{\mathcal{F}}_\rho(0, z) &= \frac{T_{L-\rho}(y)}{T_L(y)}, & \tilde{W}_\rho(L, z) &= \frac{U_{\rho-1}(y)}{zq T_L(y)}, \\ 1 - \tilde{\mathcal{F}}_L(0, z) &= \frac{T_L(y) - 1}{T_L(y)}, & \tilde{H}_{N, N/2}(z) &= \frac{T_L(y)}{D(y)} \end{aligned} \quad (48)$$

gives

$$\begin{aligned} \tilde{F}_\rho(0, z) &= \frac{T_{L-\rho}(y)}{T_L(y)} + \frac{U_{\rho-1}(y) [T_L(y) - 1]}{T_L(y) D(y)} \\ &= \frac{T_{L-\rho}(y) D(y) + U_{\rho-1}(y) [T_L(y) - 1]}{T_L(y) D(y)} \\ &= \frac{aT_{L-\rho}(y) T_L(y) + T_{L-\rho}(y) U_{L-1}(y) + U_{\rho-1}(y) T_L(y) - U_{\rho-1}(y)}{T_L(y) D(y)}. \end{aligned} \quad (49)$$

The Chebyshev identity

$$T_L(y) U_{L-\rho-1}(y) = T_{L-\rho}(y) U_{L-1}(y) - U_{\rho-1}(y)$$

then turns the numerator of Eq. (49) into

$$T_L(y) [aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)],$$

where $U_{-1} = 0$ when $\rho = L$. After cancellation of this common $T_L(y)$ factor, the expanded route gives

$$\tilde{F}_\rho(0, z) = \frac{aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)}{aT_L(y) + U_{L-1}(y)}. \quad (50)$$

Thus Eq. (45) is the corrected compact form of the expanded calculation rather than an additional assumption. In pole language, let $y = \eta_\ell$ be a zero of $T_L(y)$, corresponding to an α_ℓ -mode location in z . The common-denominator numerator in Eq. (49) is

$$N_\rho(y) = aT_{L-\rho}(y) T_L(y) + T_{L-\rho}(y) U_{L-1}(y) + U_{\rho-1}(y) T_L(y) - U_{\rho-1}(y).$$

The Chebyshev identity gives $N_\rho(y) = T_L(y) [aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)]$, so $N_\rho(\eta_\ell) = 0$. Thus the apparent $1/T_L(y)$ pole has zero residue once the terms are collected, and the collected expression has no remaining $1/T_L(y)$ factor or double pole at $T_L(y) = 0$. This is the generating-function reason why no final $t\alpha_\ell^t$ term is present.

In an Eq. (41)-style term-by-term time expansion, intermediate partial fractions may still contain expressions such as

$$\frac{s_k^t - \alpha_\ell^t}{s_k - \alpha_\ell}$$

and, in the apparent repeated-pole limit, terms proportional to $t\alpha_\ell^t$. These terms belong to the uncollected representation of the convolution. In the corrected calculation they collect into the removable $T_L(y)$ factor displayed in Eq. (49). Therefore the fully collected expanded route has the same poles as Eq. (50), namely the roots of $D(y)$, and not separate α_ℓ -poles.